

Live TV on Roku

A detailed platform capability brief: what you can deliver with confidence, what requires qualification, and practical alternatives for everything else.

Purpose. Use this document before signing statements of work or promising parity with web, mobile, or cable-style live TV products. Roku is a capable live TV platform when scope is aligned with how the device actually works.

Audience

Product, engineering, delivery, sales

Focus

Linear live TV, EPG, playback, monetization

Platform

Roku OS · SceneGraph · Native Video

Classification

Capability reference · Not a fixed estimate

Executive Summary

A professional Live TV application on Roku is **absolutely achievable**. The experience clients expect from modern TV apps — a featured live channel, a favorites row, a full electronic program guide (EPG), channel changing, and fullscreen playback — maps well to Roku’s strengths.

Where teams get into trouble is promising **web or mobile parity** without understanding platform boundaries. Roku uses a native video stack, not a browser. There is no HTML, no HLS.js, no Google IMA, no Chromecast, and no mobile-style push notifications inside your channel.

Verdict	Meaning for proposals	Examples
SUPPORTED	Safe to commit in a v1 Live TV scope	Live HLS play, EPG grid, favorites, hero preview, fullscreen player
QUALIFIED	Deliverable with caveats, backend dependency, or phased timeline	DRM, DVR, manual quality, ads, pause-live, EPG refresh
ALTERNATIVE	Different UX or technology achieves the same user goal	Inline preview instead of Picture-in-Picture (PiP); polling instead of WebSockets
NOT ON ROKU	Do not list as a Roku deliverable	Playback speed, Chromecast, mobile push, Google IMA as-is

Part I — What Roku Delivers Well

These capabilities form the core of a credible Live TV product and should anchor any v1 commitment.

1. Live channel playback (HLS)

SUPPORTED

WHAT IT IS

Streaming a broadcast or OTT live channel in real time using HTTP Live Streaming (HLS), typically delivered as an .m3u8 manifest URL from your CDN or origin.

WHY ROKU IS STRONG HERE

Roku's native **Video** node plays HLS without a third-party player SDK. The device handles segment fetching, buffering, and adaptive bitrate for live manifests when configured correctly. This is the same fundamental technology used by major FAST and live TV apps on Roku.

WHAT YOU NEED FROM THE CLIENT / BACKEND

- A playable live manifest URL per channel (or a play API that returns one)
- Token or signed-URL refresh rules if links expire
- Entitlement checks per subscription tier
- Geo / blackout policy clarity

CLIENT-SAFE COMMITMENT

Yes — live linear playback is a standard Roku deliverable, subject to stream format and DRM discovery.

2. Electronic Program Guide (EPG)

SUPPORTED

WHAT IT IS

The “TV guide” grid: channels on one axis, time on the other, with program blocks showing title, duration, and what is on now vs later. Users navigate with the remote to browse schedules and select content.

WHY ROKU IS STRONG HERE

EPG is a custom UI built in SceneGraph (groups, rectangles, labels, focus). There is no built-in EPG widget — you implement the grid — but Roku’s focus engine is designed for remote-driven 2D navigation. This is well-trodden territory for TV platforms.

WHAT YOU NEED FROM THE CLIENT / BACKEND

- Channel list with stable IDs and logos
- Schedule API: program start/end, title, metadata, optional images
- Time-window queries (e.g. now – 2h through now + 8h)
- Server time or timezone rules for the “NOW” line

CLIENT-SAFE COMMITMENT

Yes — full grid EPG is a primary engineering workstream, not a platform blocker.

3. “My Channels” / favorites row

SUPPORTED

WHAT IT IS

A horizontal row of favorite or frequently watched channels, each showing logo, current program title, time remaining, and optional progress — a quick-entry strip above the full guide.

WHY ROKU IS STRONG HERE

Horizontal scrolling rows with focus are a native TV pattern on Roku. Implementation is comparable to “continue watching” or catalog rows in any OTT app.

WHAT YOU NEED FROM THE CLIENT / BACKEND

- Favorites API or derivation from watch history
- Current program per channel (from EPG or lightweight “now playing” endpoint)

CLIENT-SAFE COMMITMENT

Yes — standard UI plus API integration.

4. Featured hero & “Watch Now”

SUPPORTED

WHAT IT IS

A large cinematic area at the top of the Live TV home screen showing the selected or featured channel: live preview or key art, LIVE badge, program title, time remaining, and a primary call-to-action to watch fullscreen.

WHY ROKU IS STRONG HERE

Hero layouts with overlaid metadata are standard on TV OTT apps. Live preview video can run in the hero while the user browses the guide below — the expected pattern for Roku live TV (see inline preview in Part II).

CLIENT-SAFE COMMITMENT

Yes — aligns with approved Live TV UX patterns.

5. Channel zap & fullscreen live player

SUPPORTED

WHAT IT IS

Changing channels from the guide (zap) and entering a dedicated fullscreen live viewing experience with appropriate controls: back, info, captions where available, and “go to live” when timeshift exists.

WHY ROKU IS STRONG HERE

Stream switching is implemented by stopping the current playback, loading a new URL, and playing — the same model used across live TV on connected devices. Fullscreen player is a dedicated screen in the app navigation stack.

LIVE VS VOD PLAYER DIFFERENCES

- Seek bar may be disabled or limited on true live edge
- “Progress” often reflects program schedule (EPG), not file duration
- Pause behavior depends on CDN capabilities (see Timeshift, Part II)

CLIENT-SAFE COMMITMENT

Yes — with live-specific player UX called out in design.

6. Automatic video quality (ABR)

SUPPORTED

WHAT IT IS

The player automatically selects the best bitrate based on network conditions — users stay on “Auto” without picking 720p or 1080p manually.

WHY ROKU IS STRONG HERE

Native HLS playback includes adaptive bitrate when fed a standard master playlist with multiple renditions. No custom player SDK required.

CLIENT-SAFE COMMITMENT

Yes — recommend Auto as default and primary quality mode for live.

Part II — Qualified Scope & Alternatives

These features are often requested. They are not “impossible,” but they must be explained accurately and scoped with dependencies.

7. Manual quality selection (mid-stream)

QUALIFIED

WHAT THE CLIENT OFTEN WANTS

A settings menu to force 1080p / 720p / 480p during playback, with a smooth switch like web HLS.js or Netflix.

ROKU REALITY

Roku does not expose fine-grained manual rendition switching during playback the way browser-based HLS libraries do. **Auto quality works well.** Manual selection typically requires stopping playback, loading a specific variant URL at the current position, and resuming — users may see a **brief rebuffer or visible glitch**.

CANNOT PROMISE

Seamless manual quality identical to web

CAN PROMISE

Auto ABR; optional manual with acceptable interruption

ALTERNATIVE FOR PROPOSALS

Ship live with **Auto only** in v1. Add manual quality in v2 if product accepts rebuffer on change.

8. Picture-in-Picture (PiP) while browsing

ALTERNATIVE

WHAT IT IS

Picture-in-Picture (PiP) means video keeps playing in a small floating window while the user does something else — for example, browsing the guide with the match still visible in a corner of the screen.

WHAT THE CLIENT OFTEN WANTS

PiP on mobile or desktop: video in a floating corner while navigating other parts of the app or the operating system.

ROKU REALITY

There is **no OS-level Picture-in-Picture (PiP)** for third-party channel apps. The channel runs full screen. Video does not persist when the user leaves the app.

THE STANDARD ALTERNATIVE: INLINE LIVE PREVIEW

Place a **Video node in the Live TV screen layout** (hero area) while the EPG occupies the lower portion. When the user moves focus across channels, swap the stream URL in that preview. This is how credible Roku live apps behave and matches typical TV Guide + preview designs.

DO NOT SAY

"Picture-in-Picture (PiP) like iPhone"

DO SAY

"Live preview while browsing the program guide"

9. Advanced DVR & cloud recording

QUALIFIED

WHAT THE CLIENT OFTEN WANTS

Record shows, series recording, storage quotas, scheduling conflicts, and a recordings library — YouTube TV / cable DVR class features.

ROKU REALITY

The Roku app provides **UI and playback**. Recording, storage, conflicts, and entitlements are **cloud/backend services** (nDVR). The TV does not record to local disk in a practical OTT model.

ARCHITECTURE (SIMPLIFIED)

- User taps Record → API schedules capture on nDVR service
- CDN stores segments → recording appears in library API
- App plays recording URL like VOD

ALTERNATIVE FOR PROPOSALS

Phase 1: Live only, optional “Start over” if catch-up URL exists.

Phase 2+: Full DVR as separate SOW tied to streaming vendor selection.

10. Ads & interactive overlays

QUALIFIED

WHAT THE CLIENT OFTEN WANTS

Same ad experience as the website: Google IMA, clickable overlays, VAST/VMAP, skip-after-5s, choose-your-ad.

ROKU REALITY

Google IMA does not run on Roku. Monetization uses **Roku Advertising Framework (RAF)** and/or **SSAI** (server-side ad insertion into the HLS stream).

SSAI

Ads stitched in manifest — app plays one URL.
Common for live. Lower client complexity.

RAF / CSAI

Client requests and plays ad assets — more control, separate integration and certification.

ALTERNATIVE FOR PROPOSALS

Do not commit to “same ad stack as web.” Commit to **ad-supported live via SSAI or RAF** as a distinct workstream.

11. Real-time EPG synchronization

ALTERNATIVE

WHAT THE CLIENT OFTEN WANTS

Instant grid updates when schedules change — WebSocket-style push to every device.

ROKU REALITY

Channels typically use **HTTP polling** (e.g. every 60–180 seconds) plus a **local device clock** to animate the “NOW” line between refreshes. This is industry-normal for TV guides.

ALTERNATIVE FOR PROPOSALS

Commit to guide accuracy within **~1 minute** and refresh on channel change / screen focus. WebSockets are optional and rarely necessary for v1.

12. DRM & premium encrypted channels

QUALIFIED

WHAT IT IS

Encrypted live streams requiring a license from a DRM server (commonly Widevine) before playback — typical for premium sports and studio content.

ROKU REALITY

Roku **supports DRM** (typically Widevine) through native player configuration. However, **web DRM setup does not copy directly**. License servers must issue Roku-compatible licenses; auth headers and token refresh must be designed for TV.

PRE-COMMITMENT REQUIREMENT

Run a **hardware proof-of-concept** with one encrypted live stream before fixing delivery dates for DRM channels.

ALTERNATIVE FOR PROPOSALS

Clear HLS + signed URLs for v1 FAST-style launch; DRM tier as Phase 2 when rights require it.

13. Timeshift, pause live & rewind

QUALIFIED

WHAT THE CLIENT OFTEN WANTS

Pause live TV for 30+ minutes, rewind 15 minutes, jump back to live — cable DVR behavior.

ROKU REALITY

Plain live HLS only buffers a **short rolling window** (often seconds to a few minutes). Extended pause and rewind require **nDVR or catch-up URLs** from the CDN. The app plays whichever URL the backend provides — Roku does not create long buffers on the device.

Capability	Without nDVR	With nDVR / catch-up
Short pause	Sometimes	Yes
Pause 30+ minutes	No	Yes
Rewind within show	No	Yes
“Back to live”	N/A	Yes (switch to live-edge URL)

IMPORTANT DISTINCTION

A program progress bar (“75 minutes left”) can be shown from **EPG schedule math** without proving long pause-live works.

Part III — Not Available on Roku

Do not list these as Roku deliverables. Offer alternatives where noted.

14. Playback speed (0.5× – 2×)

NOT ON ROKU

The native Roku Video node does **not support variable playback speed** in a reliable, platform-supported way. This is a hardware/OS limitation, not a scheduling gap.

For live TV: Rarely requested for linear channels. For catch-up recordings, clients must accept that speed control is not available on Roku unless they pursue impractical server-side workarounds.

Proposal language: Exclude from Roku scope explicitly.

15. Chromecast & cast-from-phone

NOT APPLICABLE

Roku **is the TV device**. Users launch your channel on Roku; they do not cast to Roku from a custom mobile app the way they cast to Chromecast.

Alternative: A separate companion mobile app with shared backend is a different project. Deep-linking into the channel with launch parameters is possible but is not “casting.”

16. Mobile-style push notifications

NOT ON ROKU

Roku channels do not receive FCM/APNs-style notifications when closed. Users must open the channel from the Roku home screen.

Alternatives for “remind me when show starts”:

- Email or SMS from your notification platform
- Companion mobile app (separate scope)
- In-app prompts when Live TV is already open

Client Commitment Matrix

Feature	Status	Recommendation
Live HLS playback	SUPPORTED	Commit in v1
EPG grid	SUPPORTED	Commit in v1
My Channels / favorites	SUPPORTED	Commit in v1
Hero + Watch Now	SUPPORTED	Commit in v1
Fullscreen live player	SUPPORTED	Commit in v1
Auto quality (ABR)	SUPPORTED	Commit in v1
Manual quality (smooth)	QUALIFIED	Qualify or defer to v2
Live preview while browsing guide	ALTERNATIVE	Commit; use "live preview," not Picture-in-Picture (PiP)
Cloud DVR / series record	QUALIFIED	Phase 2+; backend required
Web-identical ads (IMA)	NOT ON ROKU	SSAI or RAF — separate SOW
Chromecast	N/A	Exclude from Roku scope
Push notifications	NOT ON ROKU	Email / mobile / in-app
Real-time EPG (WebSocket)	ALTERNATIVE	Polling is standard for v1
DRM premium channels	QUALIFIED	POC before date commit
Long pause / rewind live	QUALIFIED	Requires nDVR from CDN
Playback speed	NOT ON ROKU	Exclude from scope

Suggested Phased Delivery

01

Live TV MVP

Live play · EPG · channel list · favorites row · hero · channel zap · fullscreen player · Auto quality

02

Enhanced guide & playback

Catch-up / start-over · program search · captions · improved EPG refresh · optional manual quality

03

Premium & monetization

Cloud DVR UI · DRM tiers · SSAI or RAF ads · advanced entitlements

Pre-Commitment Workshop Questions

PRODUCT

- v1 must-haves vs nice-to-haves?
- Parity target competitor?
- Hero: featured vs grid-focused?

STREAMING / BACKEND

- Clear HLS or DRM day one?
- nDVR / catch-up available?
- EPG source and refresh model?
- SSAI, RAF, or ad-free v1?

Leadership one-liner. We can deliver a strong Live TV experience on Roku aligned with modern TV Guide UX. We should not commit to web/mobile parity on playback speed, Chromecast, push, or Google IMA. DVR, long pause-live, DRM, and advanced ads require streaming infrastructure and belong in separate phases with separate estimates.